

## Project Executable Files

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 1 July 2026                                                 |
| <b>Team ID</b>       | XXXXXX                                                      |
| <b>Project Name</b>  | Smart Agricultural Production Optimization Engine(OptiCrop) |
| <b>Maximum Marks</b> | 3 Marks                                                     |

### Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

#### Step 1: Submission Checklist

| S.No | Item to Submit                                                  | Submitted (Yes / No / NA) |
|------|-----------------------------------------------------------------|---------------------------|
| 1    | Complete source code (all files and folders)                    | Yes                       |
| 2    | README / Setup Guide (instructions to run the project)          | Yes                       |
| 3    | requirements.txt / package.json / dependency file               | Yes                       |
| 4    | Database schema / seed files (if applicable)                    | NA                        |
| 5    | Environment configuration file (.env.example or similar)        | No                        |
| 6    | Deployed application URL (if hosted)                            | Yes                       |
| 7    | APK / Executable binary (if applicable for mobile/desktop apps) | NA                        |
| 8    | Dockerfile / Containerization config (if applicable)            | No                        |
| 9    | Test files and test results                                     | Yes                       |
| 10   | Demo video or walkthrough (if required)                         |                           |



## Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

OptiCrop-AI-Powered-Agricultural-Recommendation-System/

```

├── dataset/
├── models/
├── static/
├── templates/
├── screenshots/
├── testing/
├── app.py
├── predict.py
├── train_model.py
├── requirements.txt
└── README.md
  
```

## Step 3: Deployment / Access Details

| Field                              | Details                                                                                                                                                                                                 |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hosted / Deployed URL</b>       | Not Hosted (Runs on localhost)                                                                                                                                                                          |
| <b>Login Credentials (Demo)</b>    | No login required                                                                                                                                                                                       |
| <b>Platform / Hosting Provider</b> | Local Flask Application                                                                                                                                                                                 |
| <b>Repository Link</b>             | <a href="https://github.com/nithyasowreddy/OptiCrop-Smart-Agricultural-Production-Optimization-Engine">https://github.com/nithyasowreddy/OptiCrop-Smart-Agricultural-Production-Optimization-Engine</a> |
| <b>Demo Video Link</b>             | <a href="https://drive.google.com/file/d/16D96tPqICBeCJNGgZuPgQ4P_guz7yTa5/view?usp=sharing">https://drive.google.com/file/d/16D96tPqICBeCJNGgZuPgQ4P_guz7yTa5/view?usp=sharing</a>                     |

## Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Download or clone the GitHub repository.
2. Open the project in VS Code.
3. Install dependencies:

```
pip install -r requirements.txt
```

4. Run the Flask application:

```
python app.py
```

5. Open browser and visit:

```
http://127.0.0.1:5000
```

6. Enter soil parameters and click Predict.
7. The system recommends the best suitable crop.

## Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation              | Workaround / Status                  |
|------|---------------------------------------|--------------------------------------|
| 1    | Application not deployed online       | Runs successfully on localhost       |
| 2    | Prediction depends on dataset size    | Can be improved with larger datasets |
| 3    | Weather API integration not available | Future enhancement                   |